

Hands-on Lab: Secure Network Diagram

Estimated time needed: 15 minutes

Consider this scenario. You are a newly appointed IT security professional for Llobex Corporation, an EdTech startup. While the organization has grown in terms of revenue and customers, Llobex's network infrastructure has not been updated in over three years. As a result, Llobex recently experienced a significant data breach resulting in the loss of sensitive customer information. As an aftermath of the data breach, the organization has seen a drop in customers. The Board of Directors has tasked you with overhauling and enhancing the security of Llobex's network infrastructure to prevent future incidents.

The next steps will guide you through the process of determining a secure network infrastructure.

Objectives

After completing this lab, you will be able to **create** a secure network infrastructure diagram.

Prerequisites

Before you begin, make sure to complete the [Securing Network Infrastructure](#) reading.

Exercise 1: Setting up the diagramming tool

For the purposes of this exercise, you will use a web-based online diagramming tool named **Visual Paradigm** (<https://online.visual-paradigm.com/>). In this exercise, you will sign-up for a free version of the tool.

1. Navigate to <https://online.visual-paradigm.com/>.
2. Click **Sign up**.



Visual Paradigm
Online

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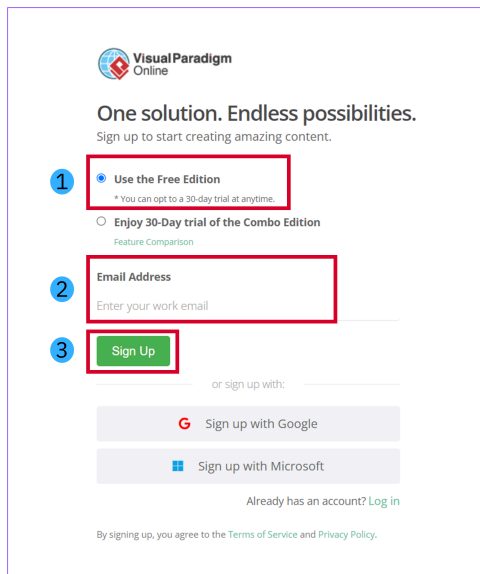
VP Online unites teams, goals and actions in one place

Visual Paradigm unites teams, goals and actions in one place

Get started for Free

3. On the Sign-up page, accept the default selection, **Use the Free Edition**.

4. Next, enter a valid email address and click **Sign Up**.



The image shows the Visual Paradigm Online sign-up page. At the top, it says "Visual Paradigm Online" with a logo. Below that, the headline is "One solution. Endless possibilities." followed by "Sign up to start creating amazing content." The page is divided into three numbered steps: 1. Selecting the edition, 2. Entering an email address, and 3. Clicking the Sign Up button. Step 1 shows two radio button options: "Use the Free Edition" (selected) and "Enjoy 30-Day trial of the Combo Edition". A red box highlights the "Use the Free Edition" option. Step 2 shows an "Email Address" input field with the placeholder text "Enter your work email". A red box highlights this field. Step 3 shows a green "Sign Up" button. A red box highlights this button. Below the input fields, there are links for "or sign up with:" followed by "Sign up with Google" and "Sign up with Microsoft". At the bottom, there is a link for "Already has an account? Log in" and a footer note: "By signing up, you agree to the Terms of Service and Privacy Policy."

Visual Paradigm Online

One solution. Endless possibilities.
Sign up to start creating amazing content.

1 ☒ Use the Free Edition
* You can opt to a 30-day trial at anytime.

☐ Enjoy 30-Day trial of the Combo Edition
Feature Comparison

2 Email Address
Enter your work email

3 Sign Up

or sign up with:

 Sign up with Google

 Sign up with Microsoft

Already has an account? [Log in](#)

By signing up, you agree to the [Terms of Service](#) and [Privacy Policy](#).

5. On the **Welcome to Visual Paradigm!** page, click the **Visit your Visual Paradigm workspace** link.

Welcome to Visual Paradigm!

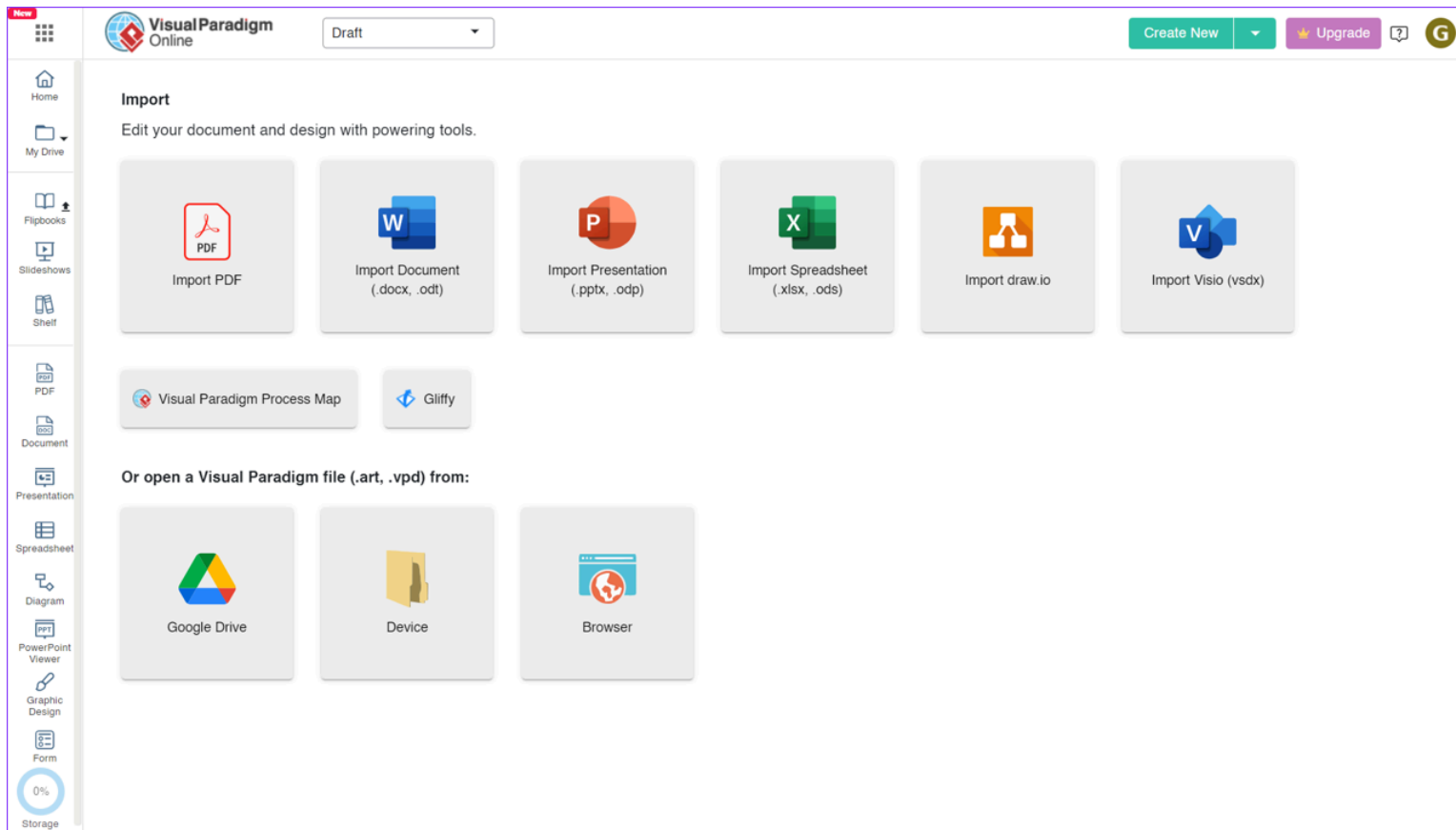
Your online workspace is ready to go. Click below to start.

[Visit your Visual Paradigm workspace](#)

To **activate your account**, just click on the link in the email we sent you and you're good to go. If you don't see the email, be sure to check your spam folder.

Enjoy!

By the way, if you have friends or coworkers who would love to use Visual Paradigm with you, invite them! There's no limit to how many people you can invite. Let's start collaborating together!



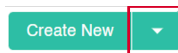
Exercise 2: Review the existing network architecture

The IT team has shared the diagram of the existing network setup. You need to analyze the diagram and make changes to ensure a secure network.

Step 1: Right-click [here](#) and then click **Save link as...** to download the file containing the existing network diagram. Save the file in the relevant folder.

Step 2: Next, import the downloaded file in **Visual Paradigm Online**.

1. In the **Visual Paradigm Online** page, click the **Create New** drop-down button.



2. Then, select **Import/Open**.

3. Next, select **Device**.

New



Visual Paradigm
Online

Draft



Home



My Drive



Flipbooks



Slideshows



Shelf



PDF

Import

Edit your document and design with powering tools.



Import PDF



Import Document
(.docx, .odt)



Visual Paradigm Process Map



Gliffy



Document



Presentation



Spreadsheet



Diagram



PowerPoint
Viewer



Graphic
Design



Form



Storage

Or open a Visual Paradigm file (.art, .vpd) from:

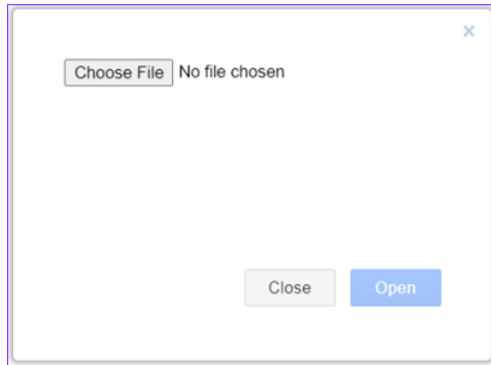


Google Drive



Device

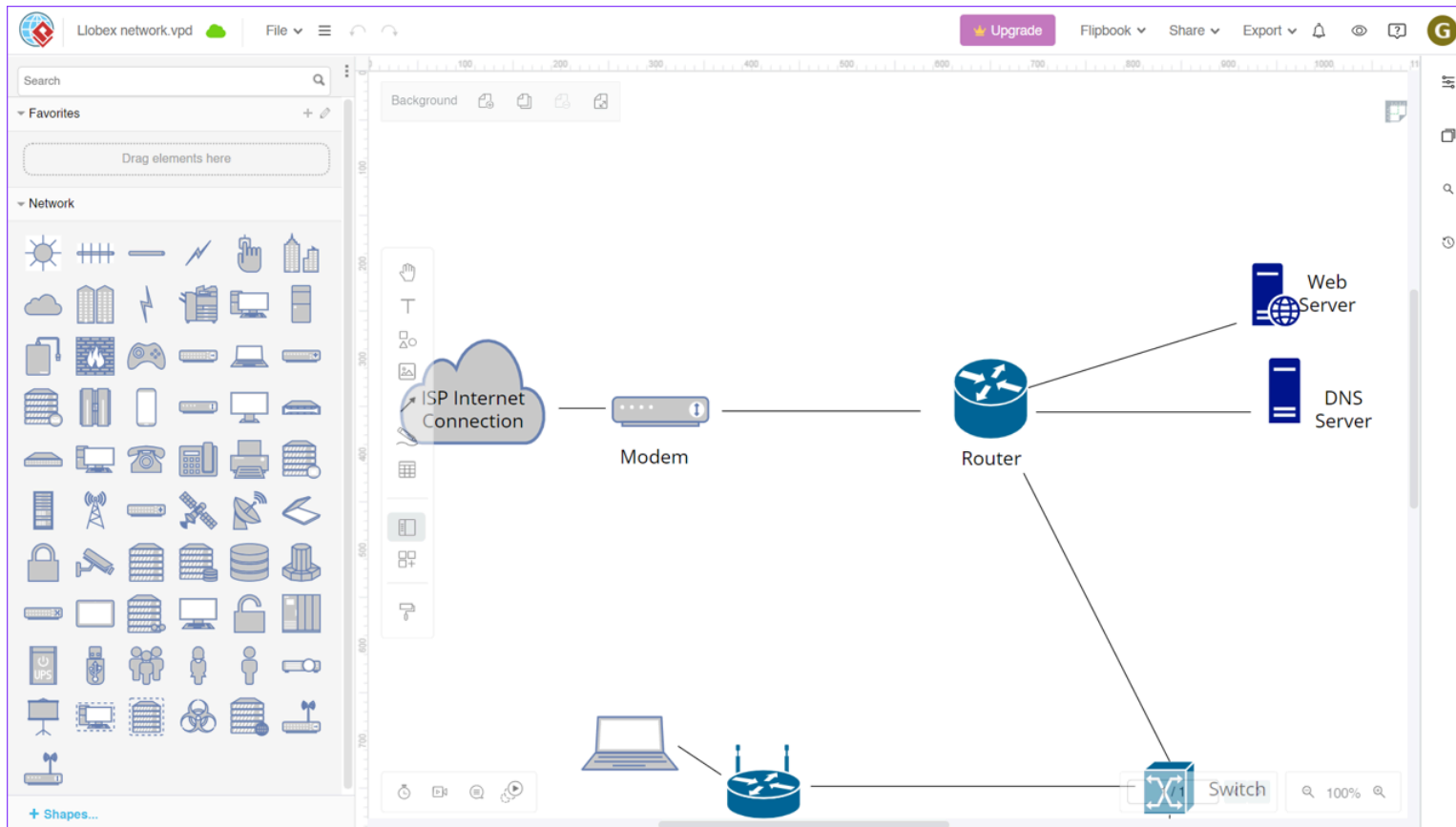
4. When prompted, click **Choose File**.



5. Navigate to the device or folder where you downloaded the file.

6. Select the file and click **Open**.

7. The network diagram with all the components in the current layout will be displayed. The network diagram consists of components, such as an Internet Service Provider (ISP), a modem, an external router, an internal switch, a wireless router, and a local area network (LAN).



8. Explore the **Visual Paradigm Online** tool.

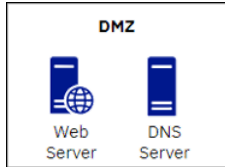
- Notice the network icons available, such as firewall and cloud, and tools, such as Text, Image, and Diagram Shapes.
- You can click a shape and drag them to the required location in the diagram.
- Right-click the shape --> click Order --> click Send to Back to move a layer backward.

Exercise 3: Update the diagram to build a secure network.

In this exercise, you will update the diagram to add components that will help build a secure network.

Step 1: Add a demilitarized zone (DMZ) to the network.

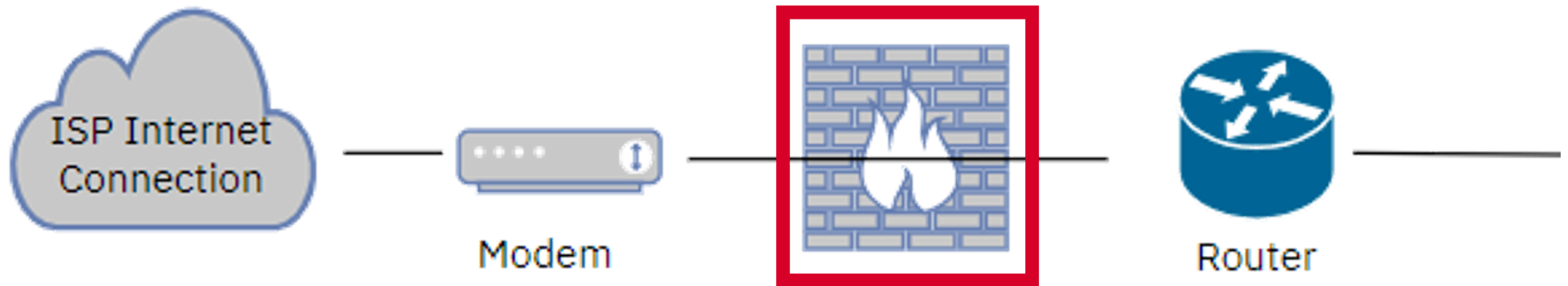
Add a DMZ to the network, which is a separate network that should include a web server and a DNS server. The DMZ should be separated from the internal LAN.



In the **Visual Paradigm Online** tool, use the tool bar on the left of the diagram to add shapes and text. Right-click the shape --> click Order --> click Send to Back to make the Web sever and DNS server visible.

Step 2: Add two firewalls to the network.

Add two firewalls to the network. One to filter external traffic and another to filter internal traffic.



In the **Visual Paradigm Online** tool, you can use the search bar to find a network component.

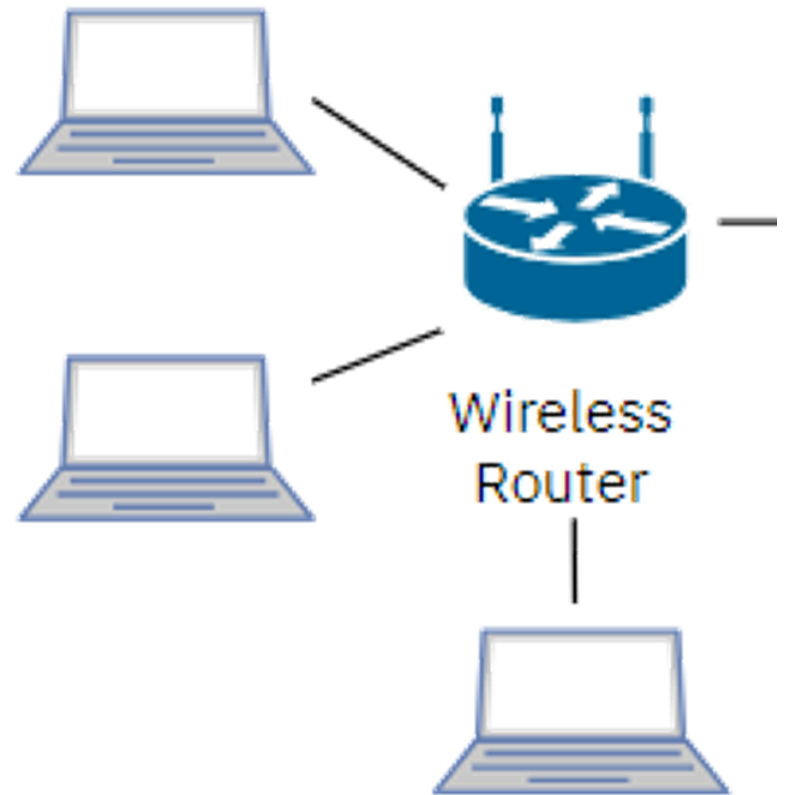
Step 3: Add an intrusion detection system (IDS).

Add an IDS to the network, which will monitor internal network traffic and provide alerts for any malicious activities detected.



Modem

Ro



In the **Visual Paradigm Online** tool, you can use the search bar to lookup a server component.

Author(s)

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